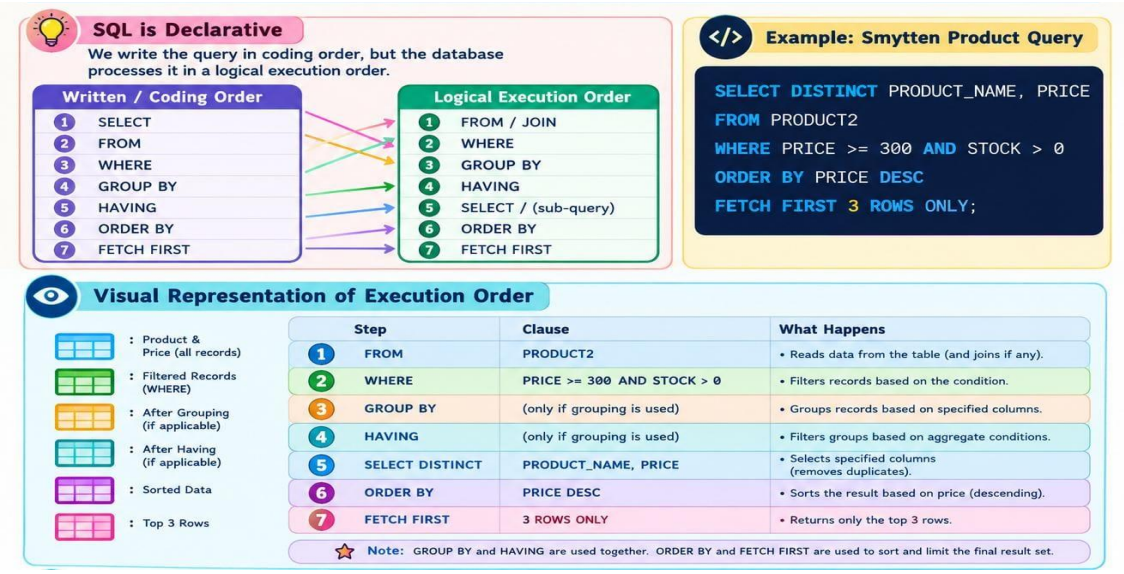
DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Gaming-Accessories E-Commerce Order Management System

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all products:

```
SQL>  SELECT * FROM Products;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                       PRICE      STOCK
------------------------------ ---------- ----------
       101 Gaming Mouse
Accessories                      799         20

       102 Gaming Keyboard
Accessories                     1499         15

       103 Gaming Headset
Audio                           1999         10


PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                       PRICE      STOCK
------------------------------ ---------- ----------
       104 Gaming Controller
Controllers                     1299          0

       105 Mouse Pad
Accessories                      299         30

       106 Gaming Chair
Furniture                       5999          5


6 rows selected.
```

Display selected product details:

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Products;

PRODUCT_ID PRODUCT_NAME                                             PRICE
---------- -------------------------------------------------- ----------
       101 Gaming Mouse                                              799
       102 Gaming Keyboard                                          1499
       103 Gaming Headset                                           1999
       104 Gaming Controller                                        1299
       105 Mouse Pad                                                 299
       106 Gaming Chair                                             5999

6 rows selected.
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

```
SQL> SELECT *
  2  FROM Products
  3  WHERE Price > 100;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       101 Gaming Mouse
Accessories                        799         20

       102 Gaming Keyboard
Accessories                       1499         15

       103 Gaming Headset
Audio                             1999         10


PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       104 Gaming Controller
Controllers                       1299          0

       105 Mouse Pad
Accessories                        299         30

       106 Gaming Chair
Furniture                         5999          5


6 rows selected.
```

Find Products with Price Less Than 1000

```
SQL> SELECT *
  2  FROM Products
  3  WHERE Price < 1000;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                          PRICE      STOCK
------------------------------ ---------- ----------
       101 Gaming Mouse
Accessories                         799         20

       105 Mouse Pad
Accessories                         299         30
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Products
  3  ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- ------------------------------------------------ ----------
       105 Mouse Pad                                           299
       101 Gaming Mouse                                        799
       104 Gaming Controller                                  1299
       102 Gaming Keyboard                                    1499
       103 Gaming Headset                                     1999
       106 Gaming Chair                                       5999

6 rows selected.
```

Descending order of price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Products
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- ------------------------------------------------ ----------
       106 Gaming Chair                                       5999
       103 Gaming Headset                                     1999
       102 Gaming Keyboard                                    1499
       104 Gaming Controller                                  1299
       101 Gaming Mouse                                        799
       105 Mouse Pad                                           299

6 rows selected.
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values

```
SQL> SELECT DISTINCT Price
  2  FROM Products
  3  ORDER BY Price ASC;

     PRICE
----------
       299
       799
      1299
      1499
      1999
      5999

6 rows selected.
```

```
SQL> SELECT DISTINCT Price
  2  FROM Products
  3  ORDER BY Price ASC;

     PRICE
----------
       299
       799
      1299
      1499
      1999
      5999

6 rows selected.

SQL> SELECT DISTINCT Product_Name
  2  FROM Products;

PRODUCT_NAME
--------------------------------------------------
Gaming Mouse
Gaming Keyboard
Gaming Headset
Gaming Controller
Mouse Pad
Gaming Chair

6 rows selected.
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT *
  2  FROM Products
  3  WHERE Price < 2000 AND Stock > 0;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                       PRICE      STOCK
------------------------------ ---------- ----------
       101 Gaming Mouse
Accessories                      799         20

       102 Gaming Keyboard
Accessories                     1499         15

       103 Gaming Headset
Audio                           1999         10


PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                       PRICE      STOCK
------------------------------ ---------- ----------
       105 Mouse Pad
Accessories                      299         30
```

```
SQL> SELECT *
  2  FROM Products
  3  WHERE NOT Category = 'Accessories';

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                       PRICE      STOCK
------------------------------ ---------- ----------
       103 Gaming Headset
Audio                           1999         10

       104 Gaming Controller
Controllers                     1299          0

       106 Gaming Chair
Furniture                       5999          5
```

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT CustomerID, CustomerName, Email, Mobile
  2  FROM Customer;

CUSTOMERID CUSTOMERNAME
---------- ------------------------
EMAIL
--------------------------------------------------------------------------------
MOBILE
--------------
       202 Madhuu
madhu@gmail.com
7820312230

       203 alia
aliaa@gmail.com
9090903451

CUSTOMERID CUSTOMERNAME
---------- ------------------------
EMAIL
--------------------------------------------------------------------------------
MOBILE
--------------

       204 Ranbir
ranbirr@gmail.com
7834032156
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

```
SQL> SELECT CustomerID, CustomerName, Age
  2  FROM Customer
  3  WHERE Age BETWEEN 20 AND 22;

CUSTOMERID CUSTOMERNAME                        AGE
---------- ------------------------- ----------
       202 Madhuu                            20
       203 alia                              21
       204 Ranbir                            22
```

```
SQL> SELECT CustomerID, CustomerName, City
  2  FROM Customer
  3  WHERE City = 'Chennai' OR City = 'trichy';

CUSTOMERID CUSTOMERNAME
---------- ------------------------
CITY
--------------------------------------------------
       203 alia
trichy
```

```
SQL> SELECT *
  2  FROM Products
  3  WHERE NOT Category = 'Accessories';

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
CATEGORY                         PRICE      STOCK
------------------------------ ---------- ----------
       103 Gaming Headset
Audio                             1999         10

       104 Gaming Controller
Controllers                       1299          0

       106 Gaming Chair
Furniture                         5999          5
```

5. Basic Business Reports

5.1. Customer Report

```
SQL> SELECT CustomerID, CustomerName, City
  2  FROM Customer
  3  ORDER BY CustomerName ASC;

CUSTOMERID CUSTOMERNAME
---------- ------------------------
CITY
--------------------------------------------------
       202 Madhuu
Coimbatore

       204 Ranbir
delhi

       203 alia
trichy
```

5.2. Product Report

```
SQL> SELECT
  2      Product_ID,
  3      Product_Name,
  4      Price
  5  FROM Products
  6  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------------- ----------
       106 Gaming Chair                                         5999
       103 Gaming Headset                                       1999
       102 Gaming Keyboard                                      1499
       104 Gaming Controller                                    1299
       101 Gaming Mouse                                          799
       105 Mouse Pad                                             299

6 rows selected.
```